Detailed Memo - MAS pages 237-239

CONSOLIDATION AND EXTENSION EXERCISE

(a) (1) $PQ = \sqrt{(x_Q - x_P)^2 + (y_Q - y_P)^2}$

$\therefore PQ = \sqrt{(4 - (-8))^2 + (14 - 9)^2}$

$\therefore PQ = 13$ units

(2) $AB = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2}$

$\therefore AB = \sqrt{(-5 - (-2))^2 + (4 - (-3))^2}$

$\therefore AB = \sqrt{58}$ units

(3) Label the points: A$(-6 ; -3)$ and B$(2 ; 1)$

$AB = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2}$

$\therefore AB = \sqrt{(2 - (-6))^2 + (1 - (-3))^2}$

$\therefore AB = 4\sqrt{5} = 8{,}94$ units

(b) (1) $x_M = \dfrac{x_C + x_D}{2}$

$\therefore x_M = \dfrac{-3+6}{2}$

$\therefore x_M = \dfrac{3}{2}$

$\therefore \mathrm{M}\left(\dfrac{3}{2}; \dfrac{5}{2}\right)$

$y_M = \dfrac{y_C + y_D}{2}$

$\therefore y_M = \dfrac{-2+7}{2}$

$\therefore y_M = \dfrac{5}{2}$

(2) $x_T = \dfrac{x_R + x_S}{2}$

$\therefore 4 = \dfrac{x_R - 2}{2}$

$\therefore 8 = x_R - 2$

$\therefore x_R = 10$

$\therefore \mathrm{R}(10\ ;\ 9)$

$y_T = \dfrac{y_R + y_S}{2}$

$\therefore 6 = \dfrac{y_R + 3}{2}$

$\therefore 12 = y_R + 3$

$\therefore y_R = 9$

(c) (1) $m_{RS} = \dfrac{y_S - y_R}{x_S - x_R}$

$\therefore m_{RS} = \dfrac{-1-5}{6-(-2)}$

$\therefore m_{RS} = \dfrac{-6}{8}$

$\therefore m_{RS} = -\dfrac{3}{4}$

(2) Label the points: A$(-3\ ;\ -6)$ and B$(1\ ;\ 4)$

$m_{RS} = \dfrac{y_B - y_A}{x_B - x_A}$

$\therefore m_{AB} = \dfrac{4-(-6)}{1-(-3)}$

$\therefore m_{AB} = \dfrac{10}{4}$

$\therefore m_{AB} = \dfrac{5}{2}$

(d) (1) $\mathrm{CD} = \sqrt{(x_D - x_C)^2 + (y_D - y_C)^2}$

$\therefore \mathrm{CD} = \sqrt{(-9-(-1))^2 + (11-4)^2}$

$\therefore \mathrm{CD} = \sqrt{113}$ units

$\therefore \mathrm{CE} = \sqrt{(x_E - x_C)^2 + (y_E - y_C)^2}$

$\therefore \mathrm{CE} = \sqrt{(6-(-1))^2 + (-4-4)^2}$

$\therefore \mathrm{CE} = \sqrt{113}$ units

$\therefore \mathrm{CD} = \mathrm{CE}$

$\therefore$ C is equidistant from D and E.

(2) Label the points: A$(-2\ ;\ -4)$; B$(2\ ;\ -10)$; C$(-1\ ;\ 11)$ and D$(3\ ;\ 5)$

$m_{AB} = \dfrac{y_B - y_A}{x_B - x_A}$

$\therefore m_{AB} = \dfrac{-10-(-4)}{2-(-2)}$

$\therefore m_{AB} = \dfrac{-6}{4}$

$\therefore m_{AB} = -\dfrac{3}{2}$

$\therefore m_{AB} = m_{CD}$

$\therefore \mathrm{AB} \parallel \mathrm{CD}$

$m_{CD} = \dfrac{y_D - y_C}{x_D - x_C}$

$\therefore m_{CD} = \dfrac{5-11}{3-(-1)}$

$\therefore m_{CD} = \dfrac{-6}{4}$

$\therefore m_{CD} = -\dfrac{3}{2}$

(3) $m_{MN} = \dfrac{y_N - y_M}{x_N - x_M}$

$\therefore m_{MN} = \dfrac{-6-(-5)}{-7-(-3)}$

$\therefore m_{MN} = \dfrac{-1}{-4}$

$\therefore m_{MN} = \dfrac{1}{4}$

$m_{MN} \times m_{PQ} = \dfrac{1}{4} \times -4 = -1$

$\therefore \mathrm{MN} \perp \mathrm{PQ}$

$m_{PQ} = \dfrac{y_P - y_Q}{x_P - x_Q}$

$\therefore m_{PQ} = \dfrac{12-4}{5-7}$

$\therefore m_{PQ} = \dfrac{8}{-2}$

$\therefore m_{PQ} = -4$

(4) Label the points: A$(-12\ ;\ 9)$; B$(-5\ ;\ 3)$ and C$(2\ ;\ -3)$

$m_{AB} = \dfrac{y_B - y_A}{x_B - x_A}$

$\therefore m_{AB} = \dfrac{3-9}{-5-(-12)}$

$\therefore m_{AB} = -\dfrac{6}{7}$

$\therefore m_{AB} = m_{BC}$

$\therefore$ A, B and C are collinear points.

$m_{BC} = \dfrac{y_C - y_B}{x_C - x_B}$

$\therefore m_{BC} = \dfrac{-3-3}{2-(-5)}$

$\therefore m_{BC} = -\dfrac{6}{7}$

(e) (1) UV $= \sqrt{17}$ units

$\therefore \sqrt{(x_V - x_U)^2 + (y_V - y_U)^2} = \sqrt{17}$

$\therefore (-1-(-5))^2 + (4-p)^2 = 17$

$\therefore 16 + p^2 - 8p + 16 - 17 = 0$

$\therefore p^2 - 8p + 15 = 0$

$\therefore (p-5)(p-3) = 0$

$\therefore p = 5$ or $p = 3$

(2) Label the points: A(p ; 5) and B(-2 ; 9)

$\therefore m_{AB} = -\frac{2}{3}$

$\therefore \frac{y_B - y_A}{x_B - x_A} = -\frac{2}{3}$

$\therefore \frac{9-5}{-2-p} = -\frac{2}{3}$

$\therefore \frac{4}{-2-p} = -\frac{2}{3}$

$\therefore -2(-2-p) = 12$

$\therefore 2p + 4 = 12$

$\therefore 2p = 8$

$\therefore p = 4$

(3) KL = KM

$\therefore \sqrt{(x_L - x_K)^2 + (y_L - y_K)^2} = \sqrt{(x_M - x_K)^2 + (y_M - y_k)^2}$

$\therefore (-6-(-3))^2 + (2-p)^2 = (-1-(-3))^2 + (7-p)^2$

$\therefore 9 + p^2 - 4p + 4 = 4 + p^2 - 14p + 49$

$\therefore 10p = 40$

$\therefore p = 4$

(4) CD || EF

$\therefore \frac{y_D - y_C}{x_D - x_C} = \frac{x_F - y_E}{x_F - x_E}$

$\therefore \frac{6-2}{6-(-4)} = \frac{-3-p}{-2-3}$

$\therefore \frac{4}{10} = \frac{-3-p}{-5}$

$\therefore 10(-3-p) = -20$

$\therefore -3-p = -2$

$\therefore -p = 1$

$\therefore p = -1$

(5) Label the points: A(-7 ; 2) ; B(p ; -6) ; C(2 ; -8) and D(10 ; -2)

AB $\perp$ CD

$\therefore m_{AB} \times m_{CD} = -1$

$\therefore \frac{-6-2}{p-(-7)} \times \frac{-2-(-8)}{10-2} = -1$

$\therefore \frac{-8}{p+7} \times \frac{6}{8} = -1$

$\therefore \frac{6}{p+7} = 1$

$\therefore p + 7 = 6$

$\therefore p = -1$

(6) $m_{GH} = m_{HI}$

$\therefore \frac{-4-(-6)}{p-(-7)} = \frac{-3-(-4)}{8-p}$

$\therefore \frac{2}{p+7} = \frac{1}{8-p}$

$\therefore p + 7 = 2(8-p)$

$\therefore p + 7 = 16 - 2p$

$\therefore 3p = 9$

$\therefore p = 3$

(f) (1) Label the points: P(-2 ; 5) ; A(2 ; $3k-1$) ; B(-5 ; $k-2$) and $k > 0$

AP = BP

$\therefore \sqrt{(2-(-2))^2 + (3k-1-5)^2} = \sqrt{(-5-(-2))^2 + (k-2-5)^2}$

$\therefore 16 + (3k-6)^2 = 9 + (k-7)^2$

$\therefore 16 + 9k^2 - 36k + 36 = 9 + k^2 - 14k + 49$

$\therefore 8k^2 - 22k - 6 = 0$

$\therefore 4k^2 - 11k - 3 = 0$

$\therefore (4k+1)(k-3) = 0$

$\therefore k = -\frac{1}{4}$ or $k = 3$

N.A.

$\therefore k = 3$

(2) CD $\perp$ EF ; $k \leq 0$

$\therefore m_{CD} \times m_{EF} = -1$

$\therefore \frac{4-2}{2k+3-k} \times \frac{-2-7}{1-(k-5)} = -1$

$\therefore \frac{2}{k+3} \times \frac{-9}{6-k} = -1$

$\therefore \frac{-18}{(k+3)(6-k)} = -1$

$\therefore (k+3)(6-k) = 18$

$\therefore 18 + 3k - k^2 - 18 = 0$

$\therefore -k^2 + 3k = 0$

$\therefore k^2 - 3k = 0$

$\therefore k(k-3) = 0$

$\therefore k = 0$ or $k = 3$

N.A.

$\therefore k = 0$

(3) $m_{AB} = m_{BC}$

$\therefore \frac{k+3-(-3)}{k+1-(-7)} = \frac{k+3-2}{k+1-3}$

$\therefore \frac{k+6}{k+8} = \frac{k+1}{k-2}$

$\therefore (k+8)(k+1) = (k+6)(k-2)$

$\therefore k^2 + 9k + 8 = k^2 + 4k - 12$

$\therefore 5k = -20$

$\therefore k = -4$

(g) Label the points: A$(x^2\ ;\ x)$; B$(5x-6\ ;\ 2)$; C$(9\ ;-3)$

$$m_{AB}=\frac{2-x}{5x-6-x^2}$$

$$\therefore m_{AB}=\frac{2-x}{-6+5x-x^2}$$

$$\therefore m_{AB}=\frac{-(x-2)}{-(x^2-5x+6)}$$

$$\therefore m_{AB}=\frac{x-2}{(x-2)(x-3)}$$

$$\therefore m_{AB}=\frac{1}{x-3}$$

$$m_{BC}=\frac{-3-2}{9-(5x-6)}$$

$$\therefore m_{BC}=\frac{-5}{9-5x+6}$$

$$\therefore m_{BC}=\frac{-5}{15-5x}$$

$$\therefore m_{BC}=\frac{-5}{-5(x-3)}$$

$$\therefore m_{BC}=\frac{1}{x-3}$$

$\therefore m_{AB}=m_{BC}$

$\therefore$ A ; B and C are collinear points.

(h) (1) $\text{PQ}=\sqrt{(-1-(-4))^2+(3-(-4))^2}$

$\therefore \text{PQ}=\sqrt{58}$

$\text{PR}=\sqrt{(-4-3)^2+(-4-(-7))^2}$

$\therefore \text{PR}=\sqrt{58}$ units

$\therefore \text{PQ}=\text{PR}$

$\therefore \triangle$ PQR is an isosceles triangle.

(2) $x_T=\frac{x_Q+x_R}{2}$

$\therefore x_T=\frac{-1+3}{2}$

$\therefore x_T=1$

$\therefore \text{T}(1\ ;-2)$

$y_T=\frac{y_Q+y_R}{2}$

$\therefore y_T=\frac{3+(-7)}{2}$

$\therefore y_T=-2$

(3) $x_{PT}=\frac{x_T+x_P}{x_T-x_P}$

$\therefore x_{PT}=\frac{-2-(-4)}{1-(-4)}$

$\therefore m_{PT}=\frac{2}{5}$

$\therefore m_{PT}\times m_{QR}=\frac{2}{5}\times-\frac{5}{2}=-1$

$\therefore$ PT $\perp$ QR

$y_{QR}=\frac{y_R+y_Q}{x_R-x_Q}$

$\therefore y_T=\frac{-7-3}{3-(-1)}$

$\therefore m_{QR}=-\frac{5}{2}$

(4) $\text{PQ}=\sqrt{58}=7{,}62$ units

$\text{PR}=\sqrt{58}=7{,}62$ units

$\text{QR}=\sqrt{(3-(-1))^2+(-7-3)^2}$

$\therefore \text{QR}=2\sqrt{29}=10{,}77$ units

Perimeter of $\triangle$ PQR $=\text{PQ}+\text{PR}+\text{QR}$

$=7{,}62+7{,}62+10{,}77$

$=26{,}01$ units

(5) $\text{QR}=2\sqrt{29}=10{,}77$ units

$\text{PT}=\sqrt{(1-(-4))^2+(-2-(-4))^2}$

$\therefore \text{PT}=\sqrt{29}=5{,}39$ units

Area of $\triangle$ PQR $=\frac{1}{2}\times b\times\perp h$

$=\frac{1}{2}\times(2\sqrt{29})\times(\sqrt{29})$

$=29$ units^2

(i) (1) BC = AD (opposite sides of a rectangle)

$\text{AD}=\sqrt{(2-(-1))^2+(-1-(-2))^2}$

$\therefore \text{AD}=\sqrt{10}$ units

$\therefore \text{BC}=\sqrt{10}=3{,}16$ units

(2) CD $\perp$ AD (angles of a rectangle)

$m_{AD}=\frac{-2-(-1)}{-1-2}$

$\therefore m_{AD}=\frac{1}{3}$

$m_{CD}\times m_{AD}=-1$

$\therefore m_{CD}\times\frac{1}{3}=-1$

$\therefore m_{CD}=-3$

(3) $m_{CD}=-3$

$\therefore \frac{y_C-(-1)}{0-2}=-3$

$\therefore \frac{y_C+1}{-2}=-3$

$\therefore y_C+1=6$

$\therefore y_C=5$

$\therefore \text{C}(0\ ;\ 5)$

(4) $x_M=\frac{-1+0}{2}=-\frac{1}{2}$

$y_M=\frac{-2+5}{2}=\frac{3}{2}$

$\therefore \text{M}\left(-\frac{1}{2}\ ;\ \frac{3}{2}\right)$

$x_M=\frac{x_B+x_D}{2}$

$\therefore -\frac{1}{2}=\frac{x_B+2}{2}$

$\therefore x_B=-3$

$\therefore \text{B}(-3\ ;\ 4)$

$y_M=\frac{y_B+y_D}{2}$

$\therefore \frac{3}{2}=\frac{y_B+(-1)}{2}$

$\therefore y_B=4$

(5) $BC = \sqrt{10}$ units

$CD = \sqrt{(2-0)^2 + (-1-5)^2}$

$\therefore CD = \sqrt{40} = 6{,}32$ units

Area of ABCD $= l \times b$

$= \sqrt{40} \times \sqrt{10}$

$= 20$ units^2

(j) (1) $x_T = \dfrac{x_L + x_N}{2}$ $\qquad y_T = \dfrac{y_L + y_N}{2}$

$\therefore 1 = \dfrac{x_L + 2}{2}$ $\qquad \therefore y_T = \dfrac{3 + (-1)}{2}$

$\therefore x_L = 0$ $\qquad \therefore y_T = 1$

$\therefore x = 0$ and $y = 1$

(2) KM $\perp$ LN (diagonals of a kite)

$\therefore m_{KM} \times m_{LN} = -1$

$m_{LM} = \dfrac{-1-3}{2-0} = -2$

$\therefore m_{KM} \times m_{LM} = -1$

$\therefore m_{KM} \times (-2) = -1$

$\therefore m_{KM} = \dfrac{1}{2}$

(k) (1) Midpoint of PR: (diagonals bisect)

$x = \dfrac{-4+2}{2}$ $\qquad y = \dfrac{-1+3}{2}$

$\therefore x = -1$ $\qquad \therefore y = 1$

$\therefore$ Midpoint of PR $= (-1\ ;\ 1)$

$x = \dfrac{x_S + x_Q}{2}$ $\qquad y = \dfrac{y_S + y_Q}{2}$

$\therefore -1 = \dfrac{x_S + (-2)}{2}$ $\qquad \therefore 1 = \dfrac{y_S - 2}{2}$

$\therefore x_S = 0$ $\qquad \therefore y_S = 4$

$\therefore$ S(0 ; 4)

(2) Midpoint of QR: (diagonals bisect)

$x = \dfrac{-2 + (-4)}{2}$ $\qquad y = \dfrac{-2+3}{2}$

$\therefore x = -3$ $\qquad \therefore y = \dfrac{1}{2}$

$\therefore$ Midpoint of QR $= \left(-3\ ;\ \dfrac{1}{2}\right)$

$x = \dfrac{x_T + x_P}{2}$ $\qquad y = \dfrac{y_T + y_P}{2}$

$\therefore -3 = \dfrac{x_T + 2}{2}$ $\qquad \therefore \dfrac{1}{2} = \dfrac{y_T + (-1)}{2}$

$\therefore x_S = -8$ $\qquad \therefore y_T = 2$

$\therefore$ T(−8 ; 2)

(3) Midpoint of PQ: (diagonals bisect)

$x = \dfrac{-2+2}{2}$ $\qquad y = \dfrac{-2-1}{2}$

$\therefore x = 0$ $\qquad \therefore y = -\dfrac{3}{2}$

$\therefore$ Midpoint of PQ $= \left(0\ ;\ -\dfrac{3}{2}\right)$

$x = \dfrac{x_V + x_R}{2}$ $\qquad y = \dfrac{y_V + y_R}{2}$

$\therefore 0 = \dfrac{x_V + (-4)}{2}$ $\qquad \therefore -\dfrac{3}{2} = \dfrac{y_V + 3}{2}$

$\therefore x_V = 4$ $\qquad \therefore y_V = -6$

$\therefore$ V(4 ; − 6)

(l) (1) Midpoint of UW: Midpoint of TV:

$x = \dfrac{-1-2}{2}$ $\qquad x = \dfrac{-4+1}{2}$

$\therefore x = -\dfrac{3}{2}$ $\qquad \therefore x = -\dfrac{3}{2}$

$y = \dfrac{4-1}{2}$ $\qquad y = \dfrac{2+1}{2}$

$\therefore y = \dfrac{3}{2}$ $\qquad \therefore y = \dfrac{3}{2}$

Midpoint $= \left(-\dfrac{3}{2}\ ;\ \dfrac{3}{2}\right)$ $\qquad$ Midpoint $= \left(-\dfrac{3}{2}\ ;\ \dfrac{3}{2}\right)$

$\therefore$ UW and TV bisect each other.

$\therefore$ TUVW is a parallelogram. (diagonals bisect)

(2) TUVW is a parallelogram. (proven)

$m_{UW} = \dfrac{-1-4}{-2-(-1)}$ $\qquad m_{TV} = \dfrac{2-1}{-4-1}$

$\therefore m_{UW} = 5$ $\qquad \therefore m_{TV} = -\dfrac{1}{5}$

$m_{UV} \times m_{TV} = 5 \times -\dfrac{1}{5} = -1$

$\therefore$ UV $\perp$ TV

$\therefore$ TUVW is a rhombus. (parm with $\perp$ diagonals)

(3) TUVW is a rhombus. (proven)

$m_{UV} = \dfrac{4-1}{-1-1}$ $\qquad m_{VW} = \dfrac{-1-1}{-2-1}$

$\therefore m_{UV} = -\dfrac{3}{2}$ $\qquad \therefore m_{VW} = \dfrac{2}{3}$

$m_{UV} \times m_{VW} = -\dfrac{3}{2} \times \dfrac{2}{3} = -1$

$\therefore \hat{V} = 90°$

$\therefore$ TUVW is a square. (rhombus with a 90° $\angle$)

(m)* (1) $AC = 5$ units

$$AC = \sqrt{(x-5)^2 + (0-3)^2} = 5$$

$$\therefore x^2 - 10x + 25 + 9 - 25 = 0$$

$$\therefore x^2 - 10x + 9 = 0$$

$$\therefore (x-9)(x-1) = 0$$

$$\therefore x = 9 \quad \text{or} \quad x = 1$$

$\therefore$ A(9 ; 0) and B(1 ; 0)

(2) Base $= AB = 9 - 1 = 8$ units

$\perp$ height $= 3$

$$\text{Area of } \Delta ABC = \frac{1}{2} \times b \times \perp h$$

$$= \frac{1}{2} \times 8 \times 3$$

$$= 12 \text{ units}^2$$

(n)* $BC = \sqrt{(k-2-(-1))^2 + (-k-(-4))^2}$

$$\therefore BC = \sqrt{(k-1)^2 + (-k+4)^2}$$

$$\therefore BC = \sqrt{k^2 - 2k + 1 + k^2 - 8k + 16}$$

$$\therefore BC = \sqrt{2k^2 - 10k + 17} \text{ units}$$

$$AC = \sqrt{(-1-5)^2 + (-4-(-1))^2}$$

$$\therefore AC = \sqrt{45} \text{ units}$$

$$\frac{BC}{AC} = \frac{1}{3}$$

$$\therefore \frac{\sqrt{2k^2 - 10k + 17}}{\sqrt{45}} = \frac{1}{3}$$

$$\therefore 3\sqrt{2k^2 - 10k + 17} = \sqrt{45}$$

$$\therefore 9\sqrt{2k^2 - 10k + 17} = 45$$

$$\therefore 18k^2 - 90k + 153 - 45 = 0$$

$$\therefore 18k^2 - 90k + 108 = 0$$

$$\therefore k^2 - 5k + 6 = 0$$

$$\therefore (k-3)(k-2) = 0$$

$\therefore k = 3$ or $k = 2$